

Transformers

Introduction to Transformers

LLMs are built out of transformers

Transformer: a specific kind of network architecture, like a fancier feedforward network, but based on attention

Attention Is All You Need

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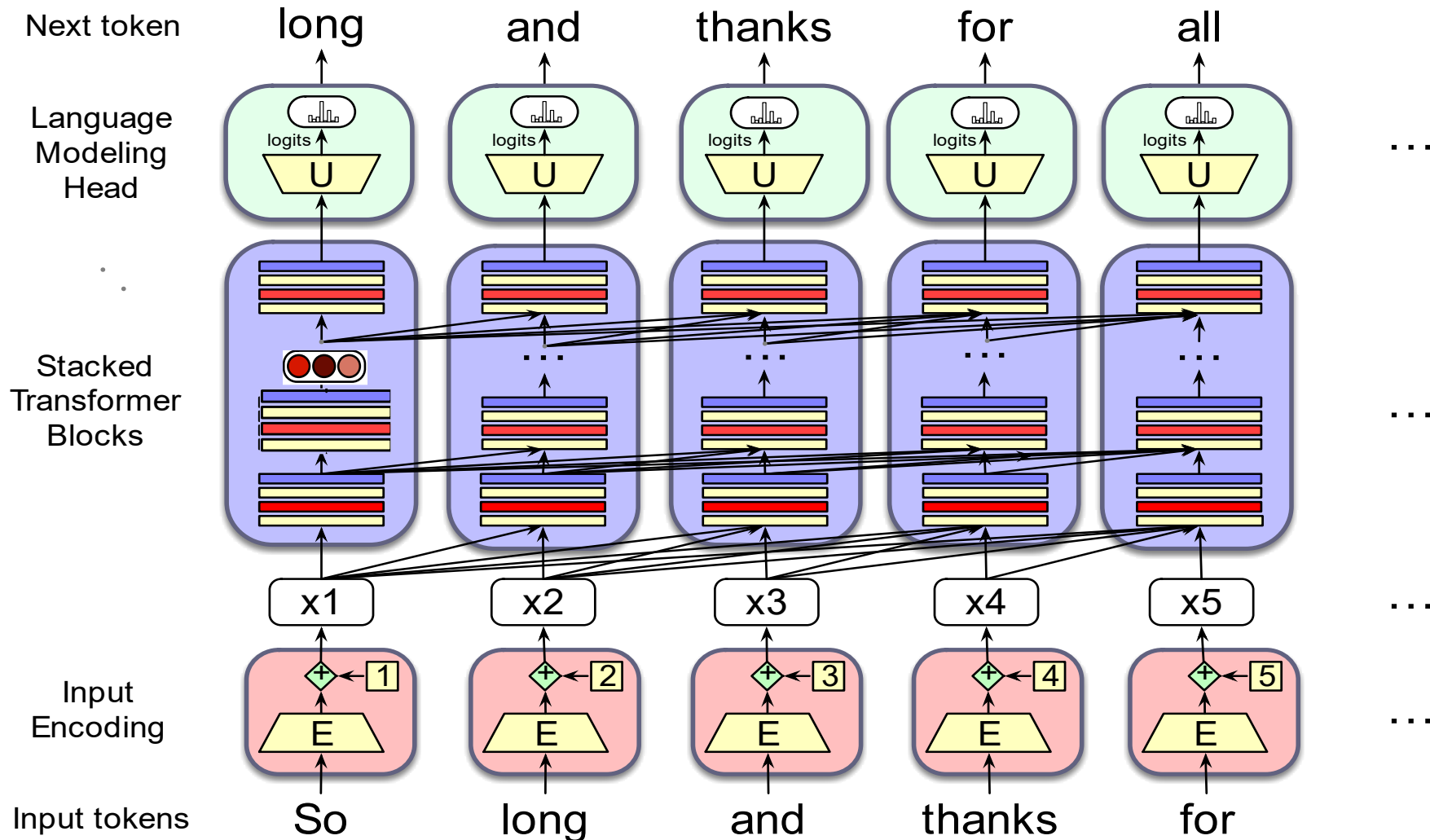
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Instead of starting with the big picture

Let's consider the embeddings for an individual word from a particular layer



Problem with static embeddings (word2vec)

They are static! The embedding for a word doesn't reflect how its meaning changes in context.

The chicken didn't cross the road because it was too tired

What is the meaning represented in the static embedding for "it"?

Contextual Embeddings

- Intuition: a representation of meaning of a word should be different in different contexts!
- **Contextual Embedding:** each word has a different vector that expresses different meanings depending on the surrounding words
- How to compute contextual embeddings?
 - **Attention**

Contextual Embeddings

The chicken didn't cross the road because it

What should be the properties of "it"?

The chicken didn't cross the road because it was too **tired**

The chicken didn't cross the road because it was too **wide**

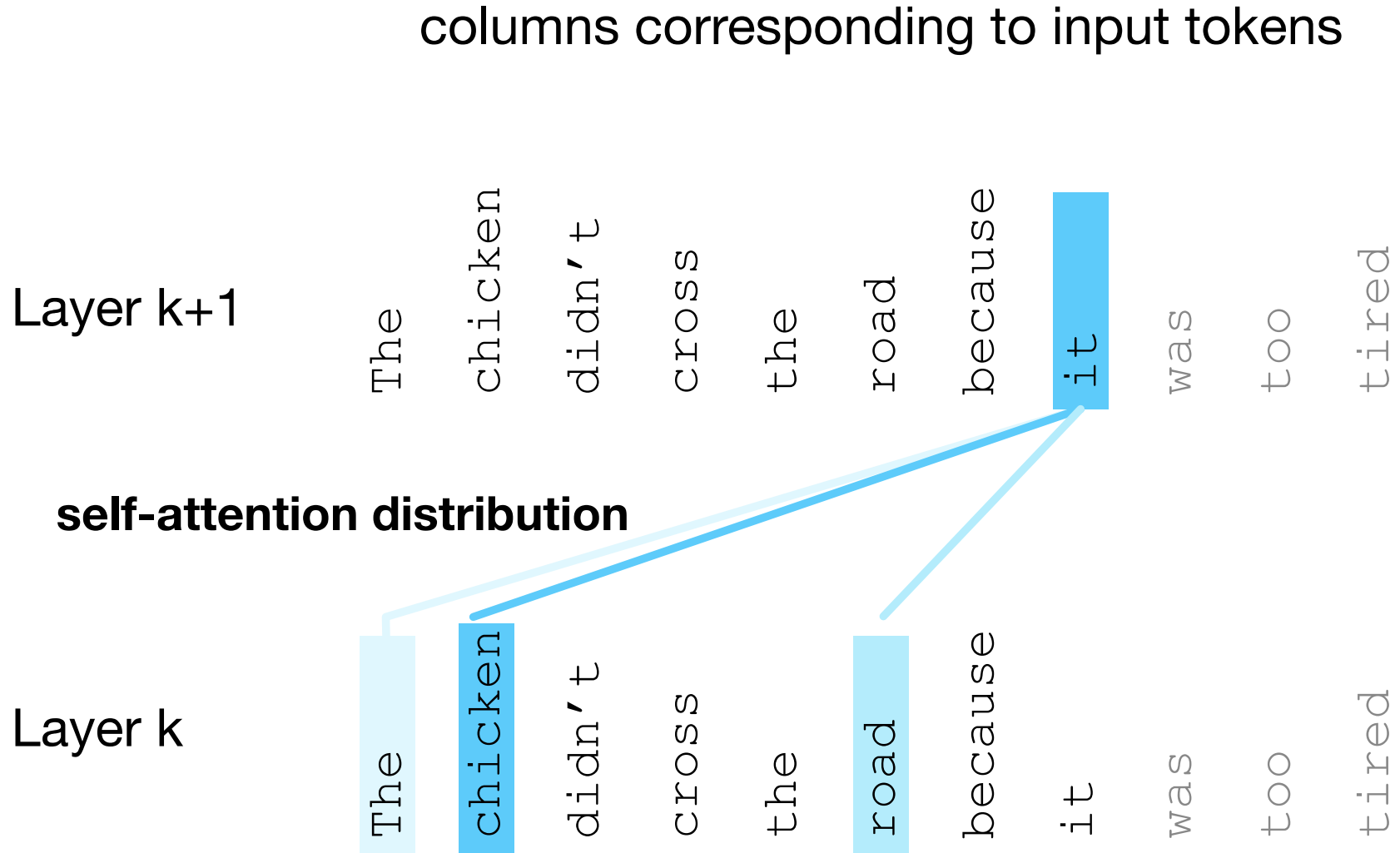
At this point in the sentence, it's probably referring to either the chicken or the street

Intuition of attention

Build up the contextual embedding from a word by selectively integrating information from all the neighboring words

We say that a word "attends to" some neighboring words more than others

Intuition of attention:

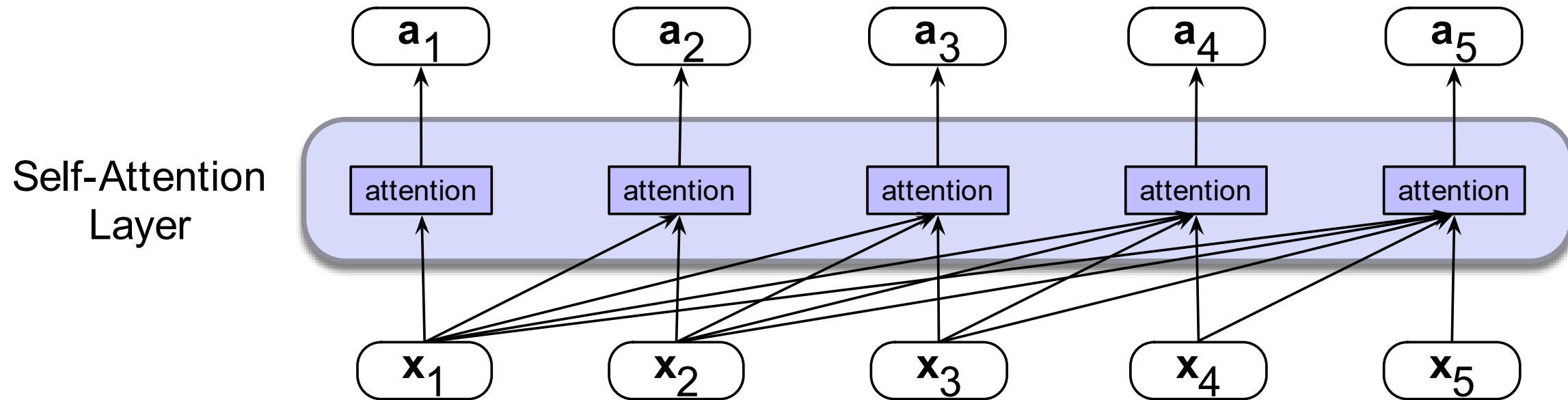


Attention definition

A mechanism for helping compute the embedding for a token by selectively attending to and integrating information from surrounding tokens (at the previous layer).

More formally: a method for doing a weighted sum of vectors.

Attention is left-to-right



Simplified version of attention: a sum of prior words weighted by their similarity with the current word

Given a sequence of token embeddings:

$$\mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_3 \quad \dots \quad \mathbf{x}_i$$

Produce: $\mathbf{a}_i$ = a weighted sum of $\mathbf{x}_1$ through $\mathbf{x}_i$

Weighted by their similarity to $\mathbf{x}_i$

$$\mathbf{a}_i = \sum_{j \leq i} \alpha_{ij} \mathbf{x}_j$$

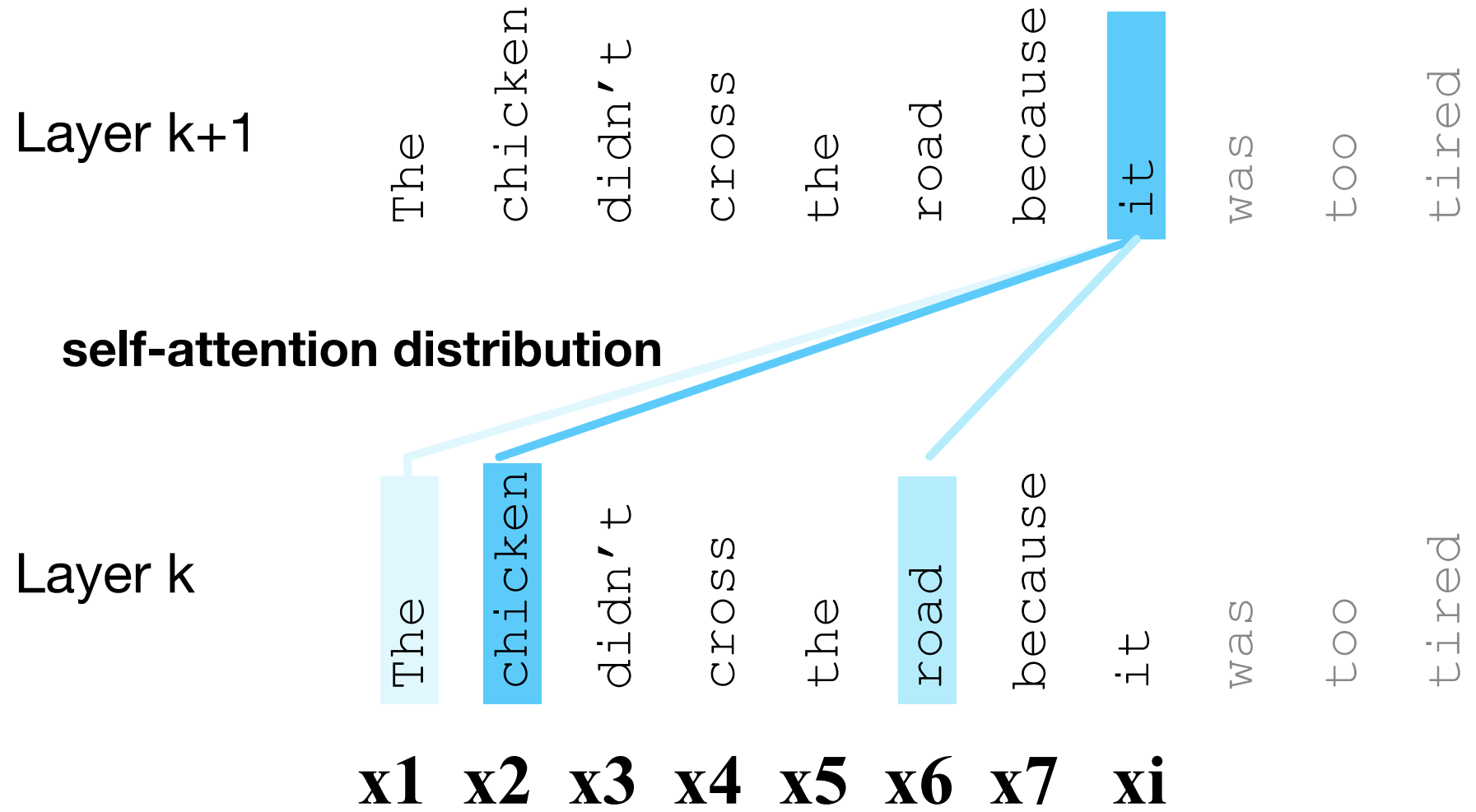
compute a_3 by computing three scores: $X_3 \cdot X_1$, $X_3 \cdot X_2$ and $X_3 \cdot X_3$, normalizing them by a softmax, and using the resulting probabilities as weights indicating each of their proportional relevance to the current position $\mathbf{i}$.

$$\text{score}(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i \cdot \mathbf{x}_j$$

$$\alpha_{ij} = \text{softmax}(\text{score}(\mathbf{x}_i, \mathbf{x}_j)) \quad \forall j \leq i$$

Intuition of attention:

columns corresponding to input tokens

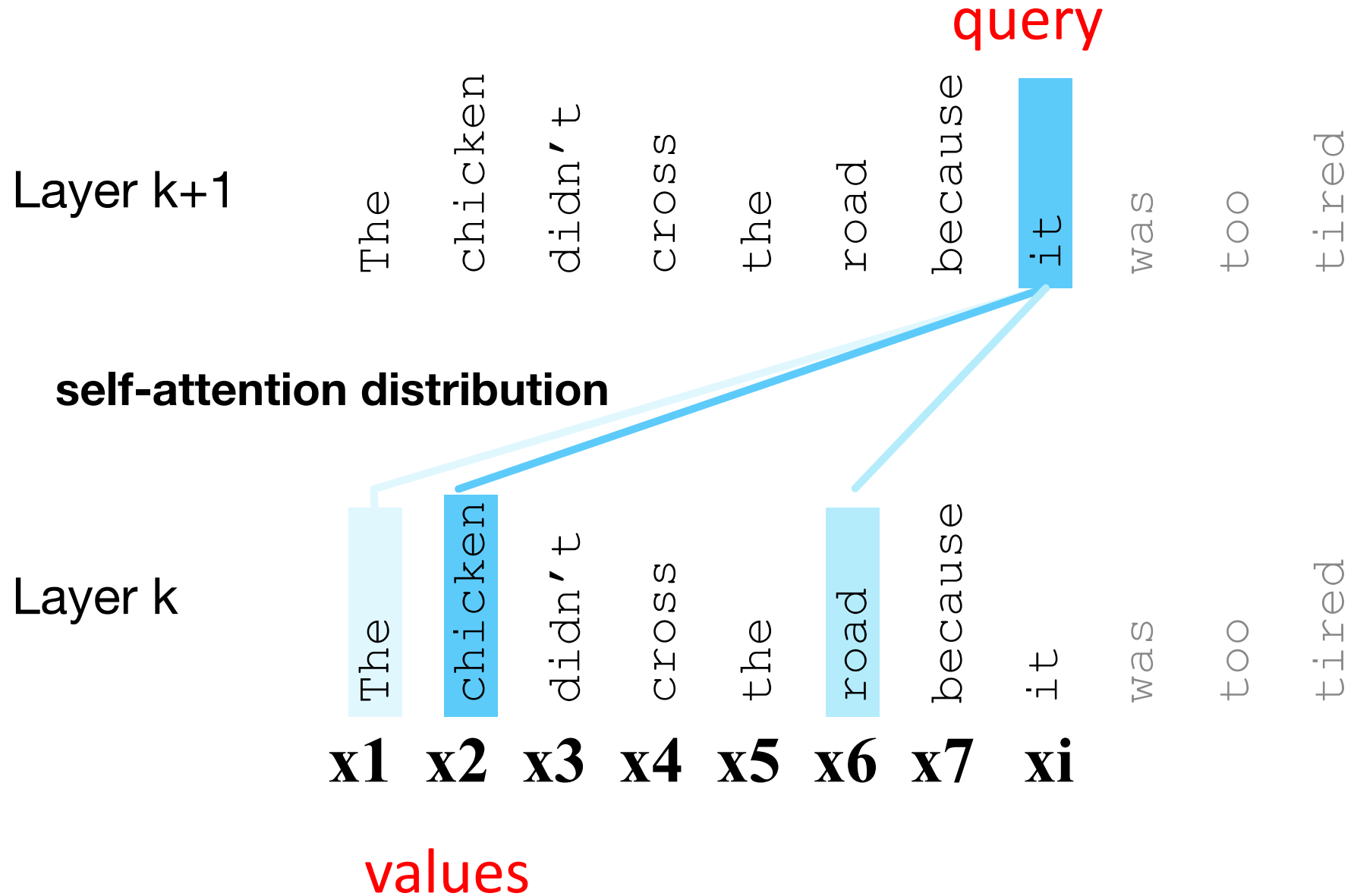


An Actual Attention Head: slightly more complicated

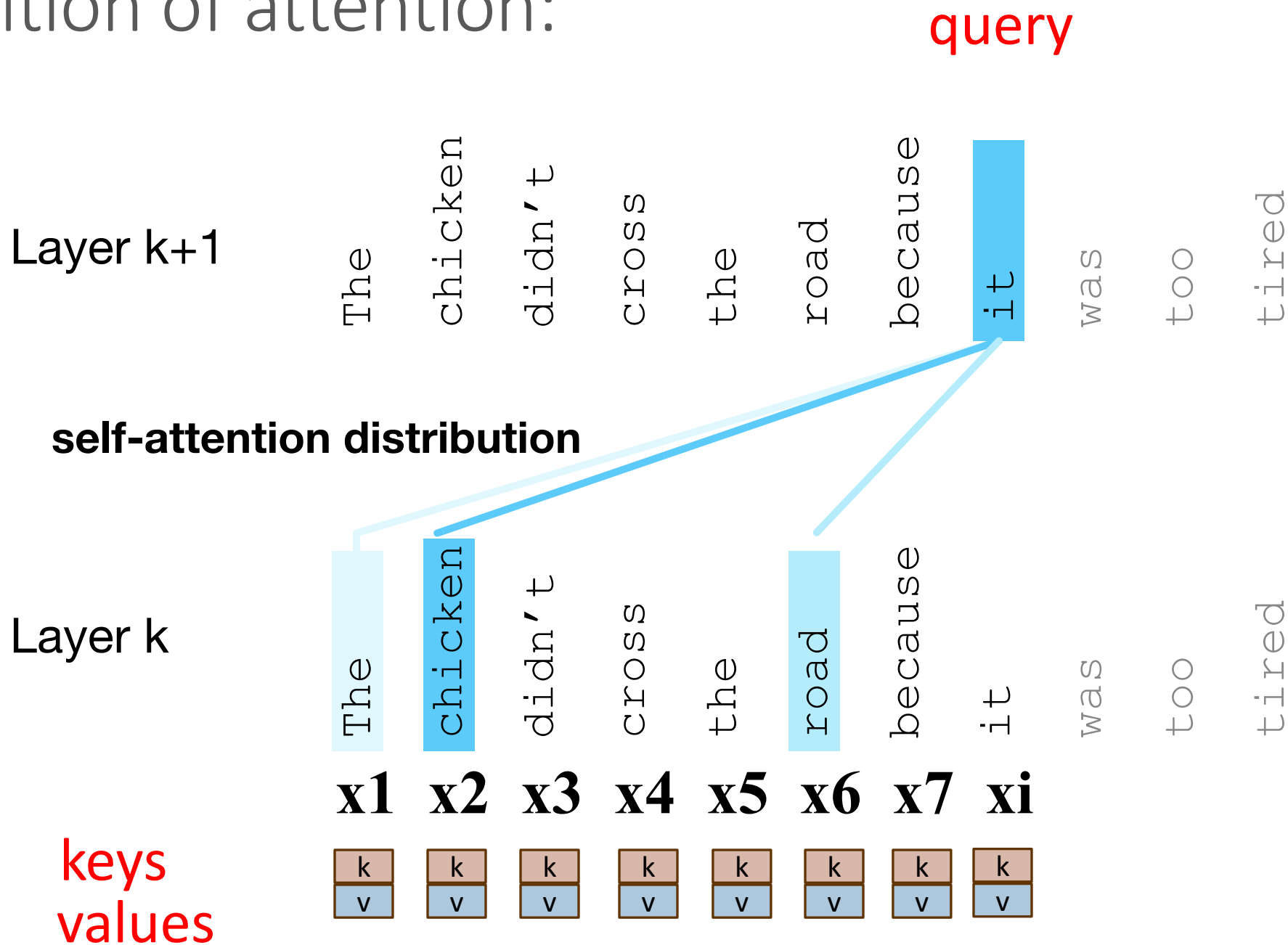
High-level idea: instead of using vectors (like x_i and x_4) directly, we'll represent 3 separate roles each vector x_i plays:

- **query**: *As the current element* being compared to the preceding inputs.
- **key**: *as a preceding input* that is being compared to the current element to determine a similarity
- **value**: a value of a preceding element that gets weighted and summed

Attention intuition



Intuition of attention:



An Actual Attention Head: slightly more complicated

We'll use matrices to project each vector $\mathbf{x}_i$ into a representation of its role as query, key, value:

- **query: $\mathbf{W}^Q$**
- **key: $\mathbf{W}^K$**
- **value: $\mathbf{W}^V$**

$$\mathbf{q}_i = \mathbf{x}_i \mathbf{W}^Q; \quad \mathbf{k}_i = \mathbf{x}_i \mathbf{W}^K; \quad \mathbf{v}_i = \mathbf{x}_i \mathbf{W}^V$$

An Actual Attention Head: slightly more complicated

Given these 3 representation of $\mathbf{x}_i$

$$\mathbf{q}_i = \mathbf{x}_i \mathbf{W}^Q; \quad \mathbf{k}_i = \mathbf{x}_i \mathbf{W}^K; \quad \mathbf{v}_i = \mathbf{x}_i \mathbf{W}^V$$

To compute similarity of current element $\mathbf{x}_i$ with some prior element $\mathbf{x}_j$

We'll use dot product between $\mathbf{q}_i$ and $\mathbf{k}_j$.

And instead of summing up $\mathbf{x}_j$, we'll sum up $\mathbf{v}_j$

Final equations for one attention head

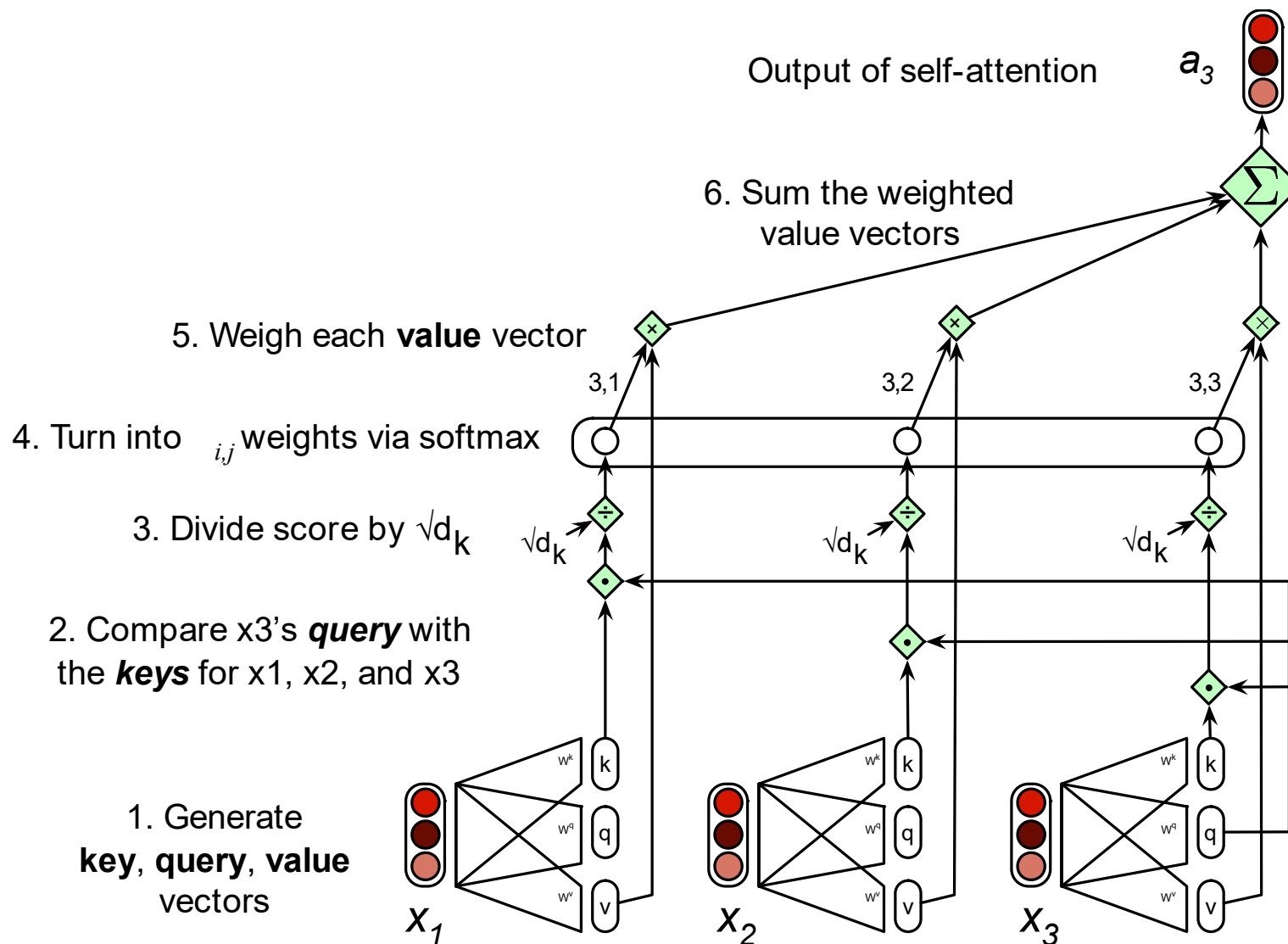
$$\mathbf{q}_i = \mathbf{x}_i \mathbf{W}^Q; \quad \mathbf{k}_j = \mathbf{x}_j \mathbf{W}^K; \quad \mathbf{v}_j = \mathbf{x}_j \mathbf{W}^V$$

$$\text{score}(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d_k}}$$

$$\alpha_{ij} = \text{softmax}(\text{score}(\mathbf{x}_i, \mathbf{x}_j)) \quad \forall j \leq i$$

$$\mathbf{a}_i = \sum_{j \leq i} \alpha_{ij} \mathbf{v}_j$$

Calculating the value of a_3



Actual Attention: slightly more complicated

- Instead of one attention head, we'll have lots of them!
- Intuition: each head might be attending to the context for different purposes
 - Different linguistic relationships or patterns in the context

$$\mathbf{q}_i^c = \mathbf{x}_i \mathbf{W}^{\mathbf{Q}^c}; \quad \mathbf{k}_j^c = \mathbf{x}_j \mathbf{W}^{\mathbf{K}^c}; \quad \mathbf{v}_j^c = \mathbf{x}_j \mathbf{W}^{\mathbf{V}^c}; \quad \forall c \quad 1 \leq c \leq h$$

$$\text{score}^c(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{q}_i^c \cdot \mathbf{k}_j^c}{\sqrt{d_k}}$$

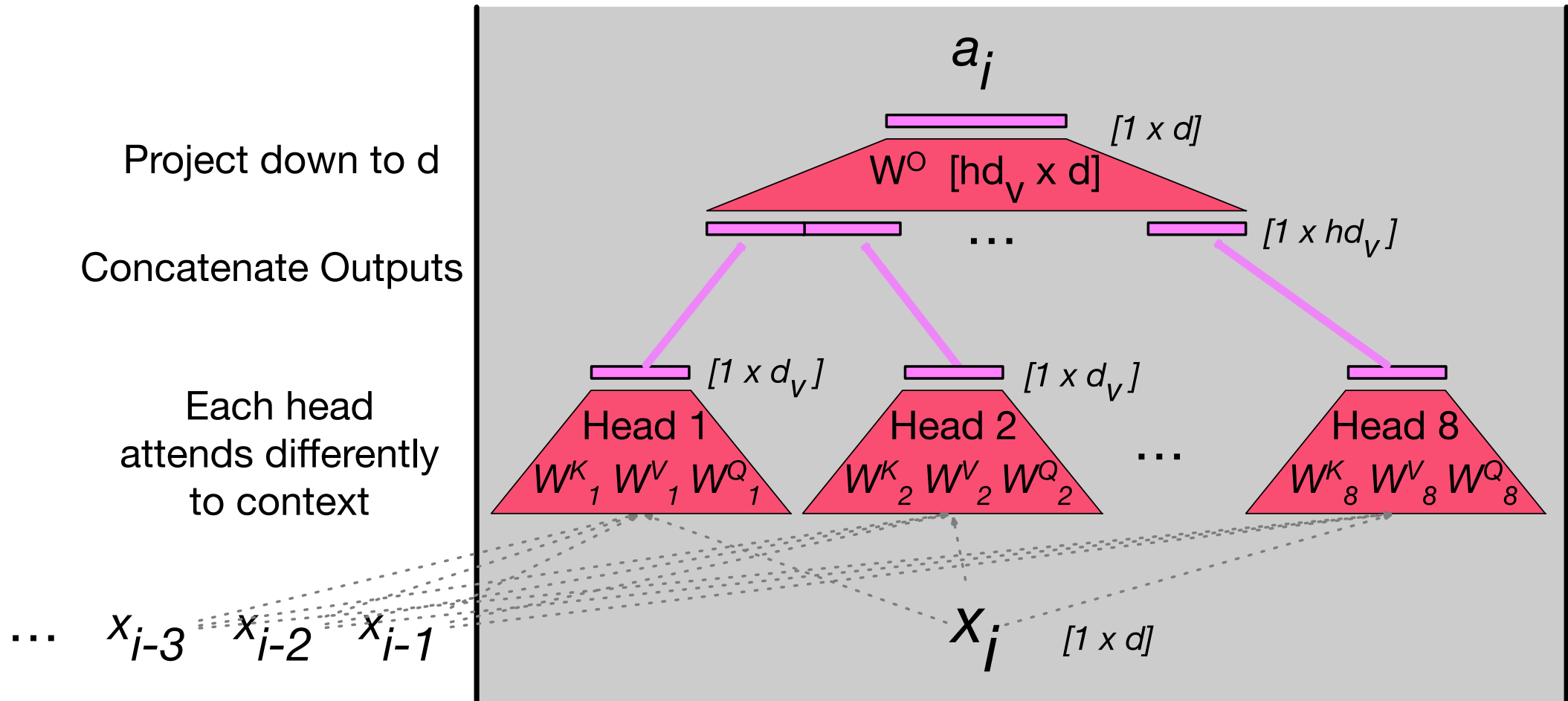
$$\alpha_{ij}^c = \text{softmax}(\text{score}^c(\mathbf{x}_i, \mathbf{x}_j)) \quad \forall j \leq i$$

$$\text{head}_i^c = \sum_{j \leq i} \alpha_{ij}^c \mathbf{v}_j^c$$

$$\mathbf{a}_i = (\text{head}^1 \oplus \text{head}^2 \dots \oplus \text{head}^h) \mathbf{W}^O$$

$$\text{MultiHeadAttention}(\mathbf{x}_i, [\mathbf{x}_1, \dots, \mathbf{x}_N]) = \mathbf{a}_i$$

Multi-head attention



Summary

Attention is a method for enriching the representation of a token by incorporating contextual information

The result: the embedding for each word will be different in different contexts!

Contextual embeddings: a representation of word meaning in its context.

We'll see in the next lecture that attention can also be viewed as a way to move information from one token to another.

References

- <https://www.youtube.com/watch?v=KMHkbXzHn7s>
- <https://www.youtube.com/watch?v=bCz4OMemCcA&t=1243s>
- <https://www.youtube.com/watch?v=ZhAz268Hdpw&t=1439s>